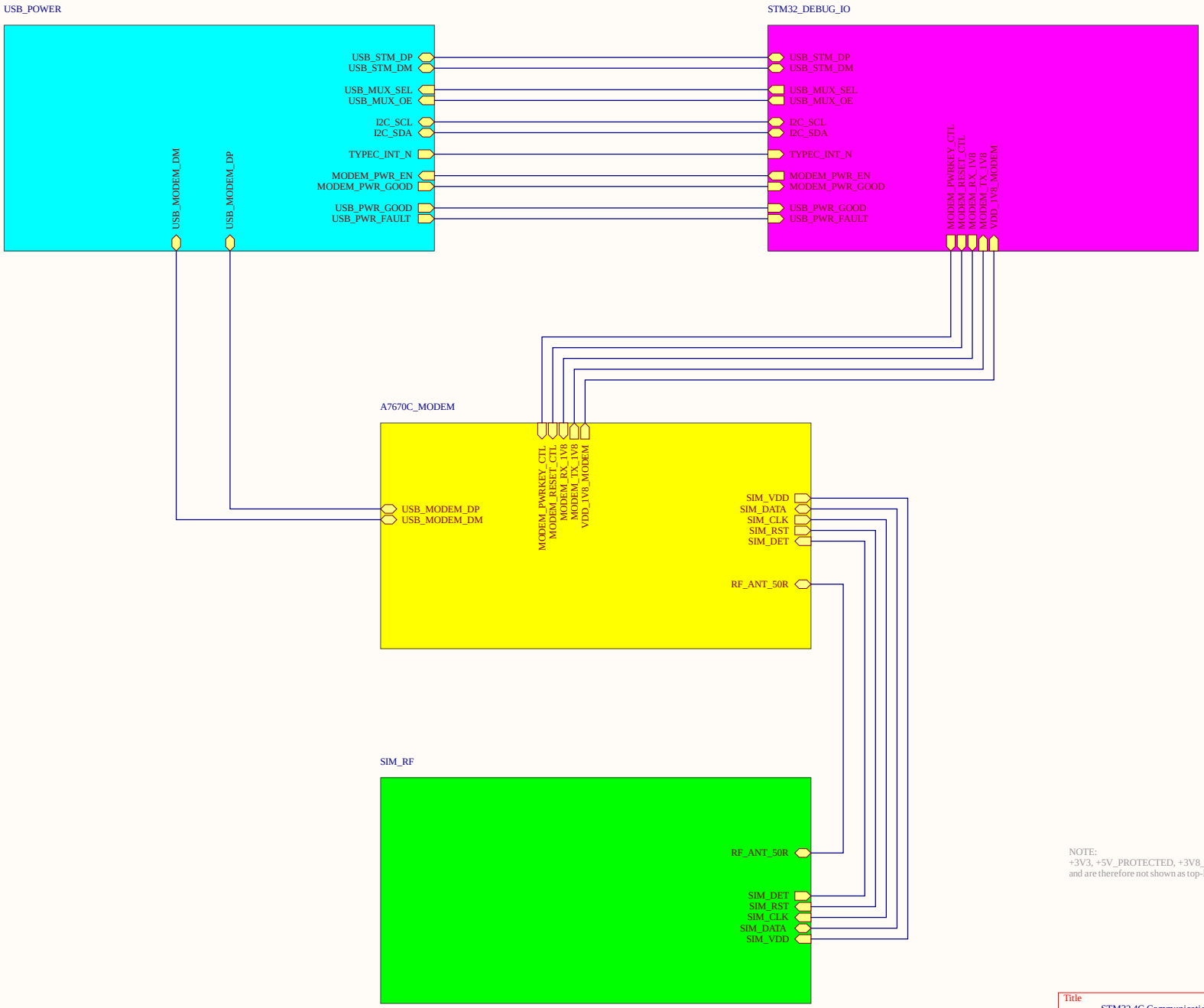
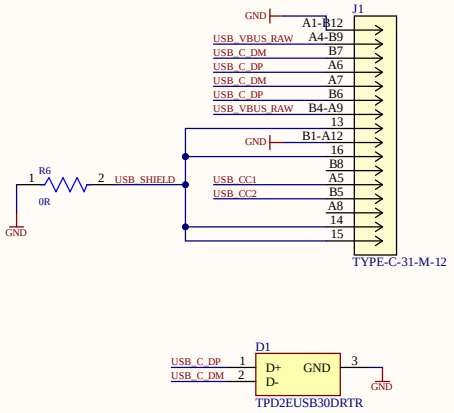


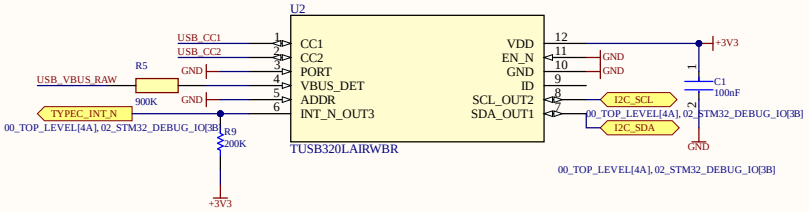
NOTE:
+3V3, +5V_PROTECTED, +3V8_MODEM and GND are global power nets and are therefore not shown as top-level hierarchical connections.

Title: STM32 4G Communication Board – System Overview		
Size: A3	Number:	Revision: A
Date: 10/24/2026		Sheet of
File: C:\Users\..\00_TOP_LEVEL.SchDoc		Drawn By: Nethmi Arnold

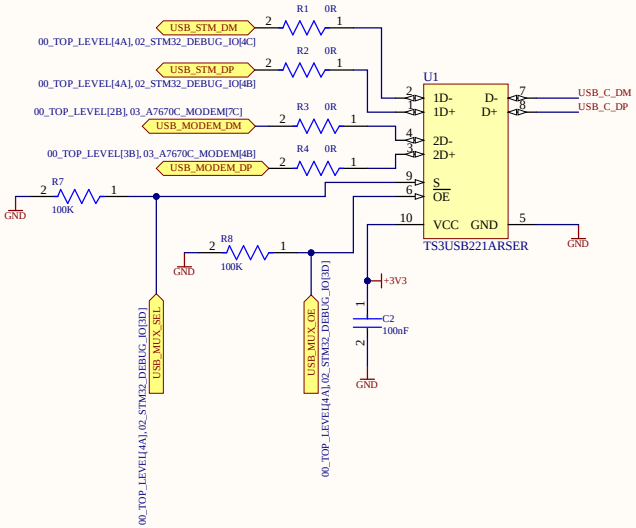
USB-C RECEPTACLE & ESD PROTECTION



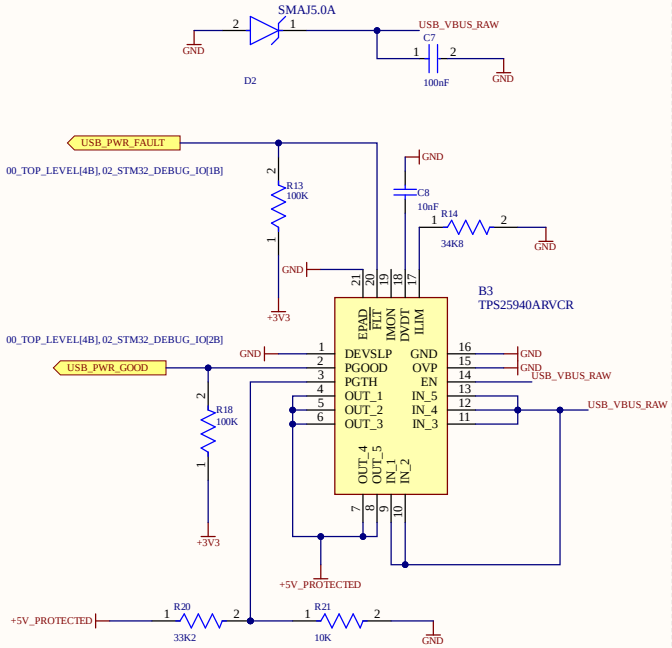
USB TYPE-C CC



USB 2.0 DATA MUX

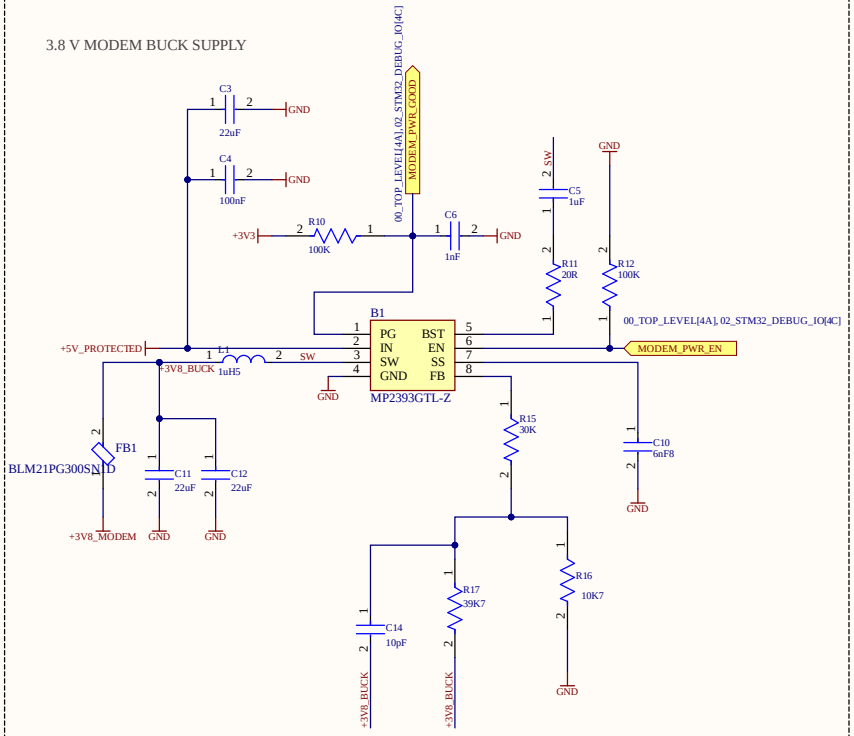


USB VBUS TRANSIENT PROTECTION & USB POWER eFUSE / 5 V PROTECTION



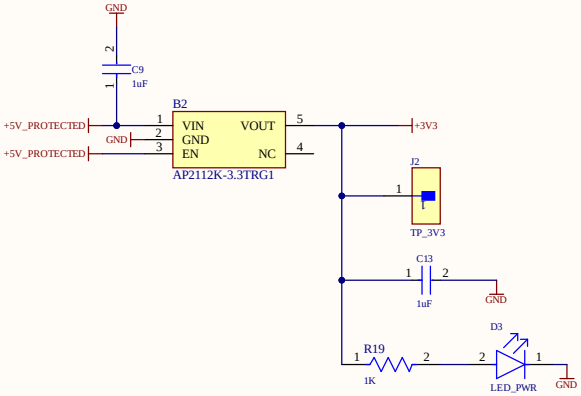
NOTES: USB power-good/fault signals are monitored by the STM32.

3.8 V MODEM BUCK SUPPLY

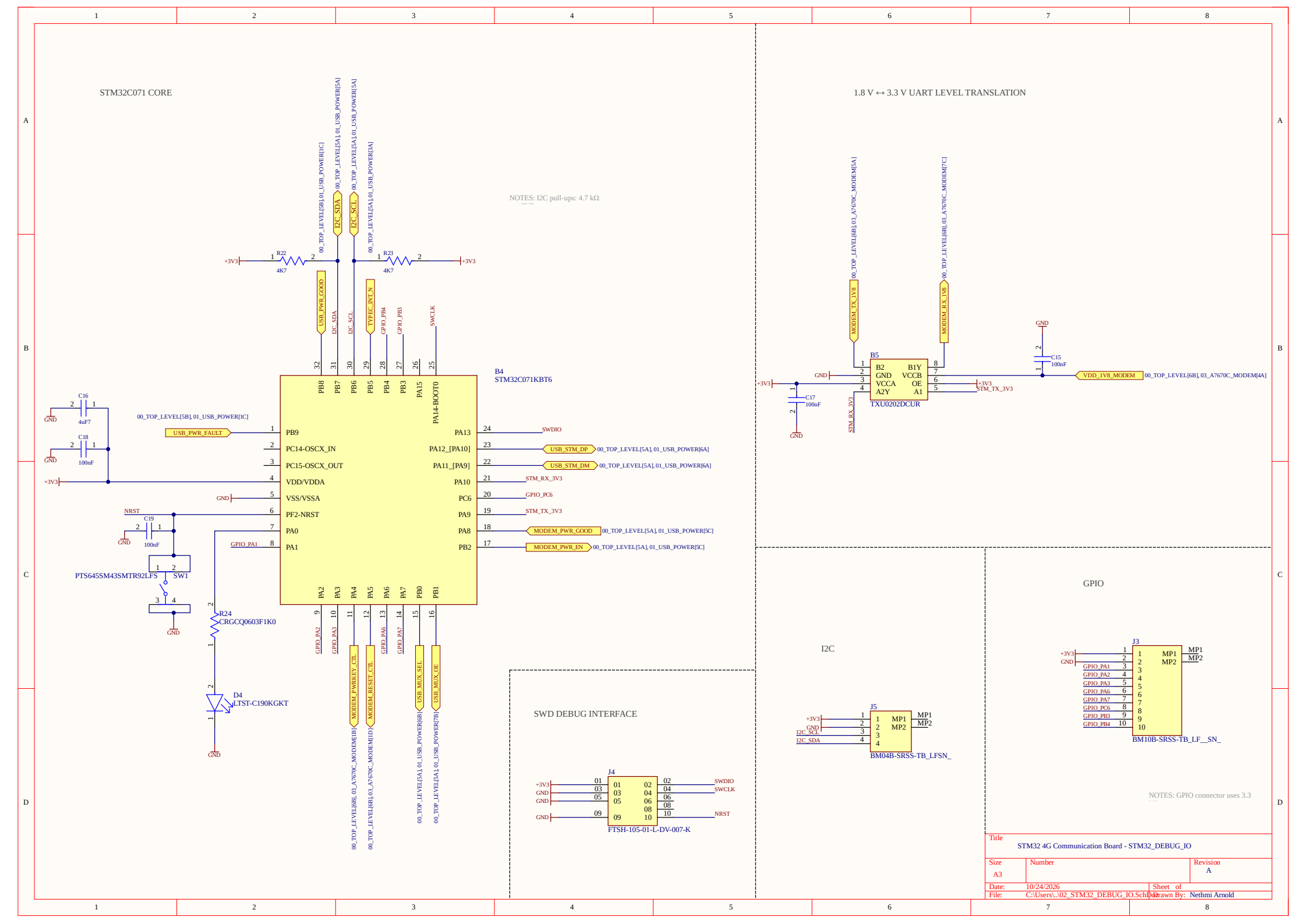


NOTES: +3V8_MODEM supplies the A7670C-LASE.

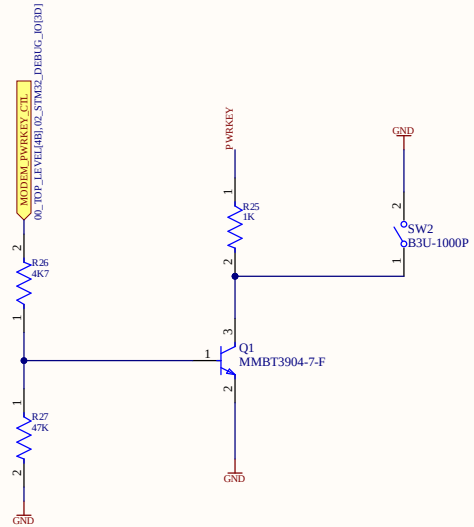
3.3 V LOGIC SUPPLY



Title	STM32 4G Communication Board - USB_POWER	
Size: A3	Number	Revision: A
Date: 10/24/2026		Sheet of
File: C:\Users\..\01_USB_POWER.SchDoc		Drawn By: Nethmi Arnold

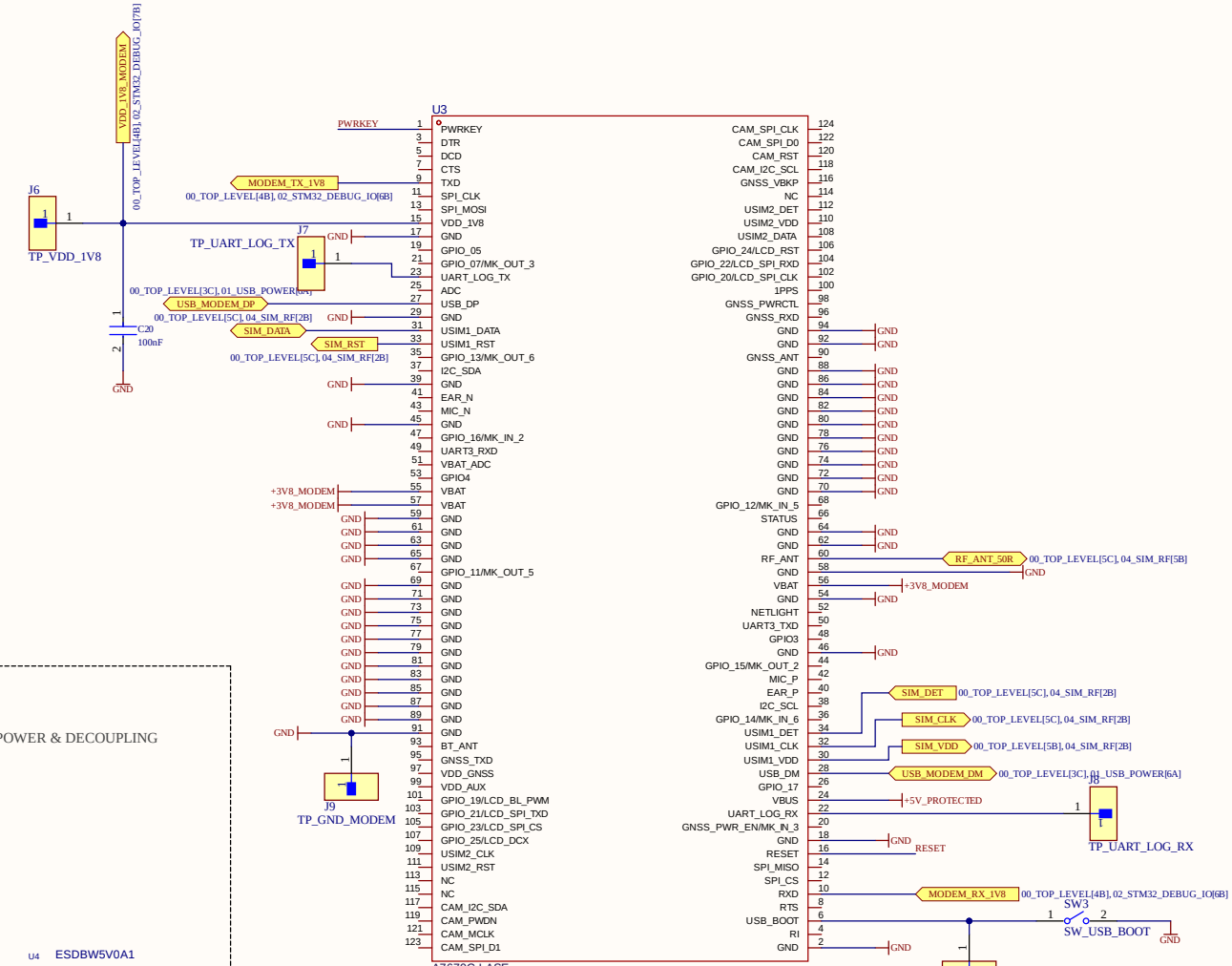

STM32C071 CORE
1.8 V ↔ 3.3 V UART LEVEL TRANSLATION
NOTES: I2C pull-ups: 4.7 kΩ
B4
STM32C071KBT6
PB8 PB7 PB6 PB5 PB4 PB3 PA15 PA14-BOOT0
PA2 PA3 PA4 PA5 PA6 PA7 PB0 PB1
PB9
PC14-OSCX_IN
PC15-OSCX_OUT
VDD/VDDA
VSS/VSSA
PF2-NRST
PA0
PA1
PA13
PA12_[PA10]
PA11_[PA9]
PA10
PC6
PA9
PA8
PB2
USB_PWR_FAULT
SWDIO
USB_STM_DP
USB_STM_DM
STM_RX_3V3
GPIO_PC6
STM_TX_3V3
MODEM_PWR_GOOD
MODEM_PWR_EN
I2C_SDA
I2C_SCL
USB_PWR_GOOD
TYPEC_INT_N
GPIO_PB4
GPIO_PB3
SWCLK
NRST
GPIO_PA1
GPIO_PA2
GPIO_PA3
MODEM_PWRKEY_CTL
MODEM_RESET_CTL
USB_MUX_SEL
USB_MUX_OE
R22
4K7
R23
4K7
C16
4uF7
C18
100nF
C19
100nF
SW1
PTS645SM43SMTR92LFS
R24
CRGCQ0603F1K0
D4
LTST-C190KGKT
B5
TXU0202DCUR
B2
GND
VCCA
A2Y
B1Y
VCCB
OE
A1
C15
100nF
C17
100nF
MODEM_TX_1V8
MODEM_RX_1V8
VDD_1V8_MODEM
STM_TX_3V3
STM_RX_3V3
I2C
J5
BM04B-SRSS-TB_LFSN_
SWD DEBUG INTERFACE
J4
FTSH-105-01-L-DV-007-K
GPIO
J3
BM10B-SRSS-TB_LF__SN_
NOTES: GPIO connector uses 3.3
Title
STM32 4G Communication Board - STM32_DEBUG_IO
Size
A3
Number
Revision
A
Date:
10/24/2026
Sheet of
File:
C:\Users\..\02_STM32_DEBUG_IO.SchDoc
Drawn By:
Nethmi Arnold

PWRKEY CONTROL



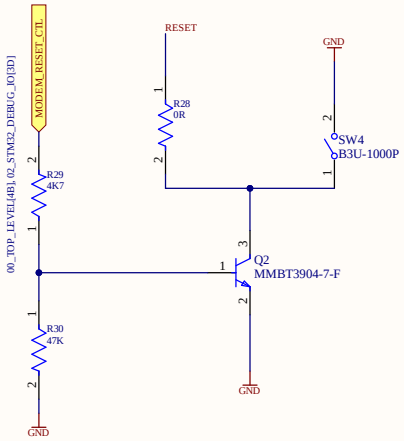
NOTES: PWRKEY and RESET are active-low.
STM32 controls them through NPN transistors.

A7670C-LASE MODEM CORE

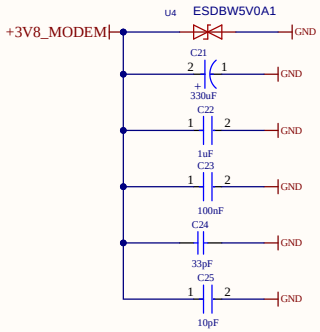


USB_BOOT:
Recovery / forced-download function only.
Normally leave released.

RESET CONTROL

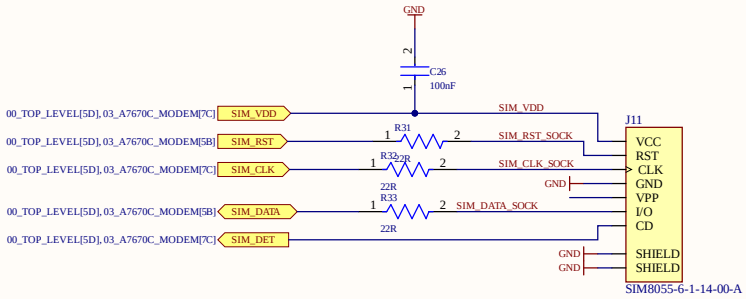


MODEM VBAT POWER & DECOUPLING

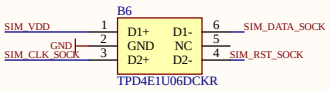


Title	STM32 4G Communication Board - A7670C_MODEM	
Size: A3	Number	Revision: A
Date: 10/24/2026	Sheet of	
File: C:\Users\..\03_A7670C_MODEM.SchDoc	Drawn By: Nethmi Arnold	

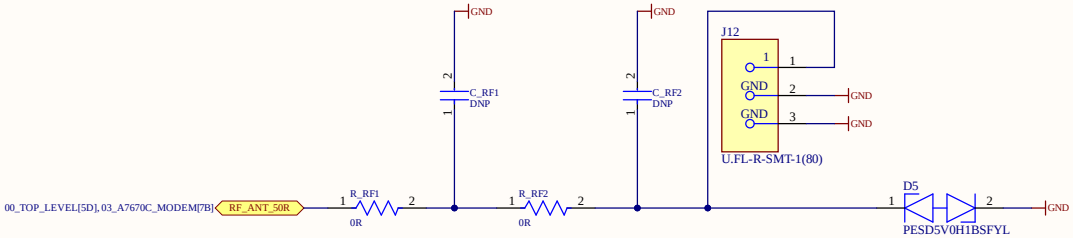
NANO-SIM INTERFACE



SIM ESD PROTECTION



LTE RF MATCHING & U.FL



NOTES:

LTE RF LAYOUT:
50 Ω controlled impedance.

DEFAULT RF MATCHING:
R_RF1 = 0 Ω
R_RF2 = 0 Ω
C_RF1 = DNP
C_RF2 = DNP

Final matching values determined by RF measurement.

Title: STM32 4G Communication Board - SIM_RF		
Size: A3	Number:	Revision: A
Date: 10/24/2026		Sheet of
File: C:\Users\..\04_SIM_RF.SchDoc		Drawn By: Nethmi Arnold